An underactuated tendon-driven gripper with variable stiffness for deformable agri-food objects

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***Abstract*— Grasping and manipulation of deformable objects are challenging problems in the food-processing industry. Food objects vary wildly in shape, size, mass, and stiffness. We present the design and control of an underactuated tendon-driven finger-based gripper with as key features high adaptability and variable joint stiffness. We first present a kinematic analysis of tendon routing in the context of grasp force and adjustable stiffness, as well as a control strategy to leverage these features. We then present a proof-of-concept prototype, which is validated in experiment. The results show that stable grasps can be achieved adaptively, and that grasp force and joint stiffness can be regulated independently for fragile objects.**

I. Introduction

Grasping and manipulation of objects remains one of the most challenging problems in robotics. These problems are especially prominent in automating the agri-food industry; food products can vary wildly in size, shape, mass, and stiffness, and are delicate, easily bruised, adhesive and/or slippery. To achieve successful grasping with minimal risk of damage, the agri-food industry employs a huge variety of specialised grippers, using electric, pneumatic, or other actuation types [1]. These grippers are typically highly specialised for a single class of objects, and used in interchangeable units to deal with inter-product class variation.

Compliant grippers benefit from adaptability to object shape while grasping. Soft robotics based on the fin-ray effect [2] or layer jamming [3] have shown decreased risk of damage, high adaptability to off-centre gripping and compliance to external force, but can not bear much load nor be controlled precisely. Similar limitations are present in underactuated rigid-body-based passive-compliance grippers. These grippers utilise rigid body links in combination with joint compliance [4]–[9] or flexible joints [10] to return to the initial position and passively adapt to the object's shape.

The human hand is marvel in its complexity, capability and robustness, and therefore well represented as an inspiration for gripper design. The muscles are often imitated by tendons routed inside the phalanges, to decrease the gripper's size and weight by locating their actuators remotely from their joints [11]. The 5-finger Shadow Hand [12] showed comparable capabilities to the human hand, even including in-hand manipulation, but with high mechanical and control complexity. A less complex implementation featuring in-hand manipulation is shown in the 3/4-finger HERI II hand [8]. The Pisa/IIT SoftHand [9], a 5-finger gripper with passive compliance, employs only two actuators and a single tendon, resulting in two synergies and lower complexity. Muscle synergies have been proposed [13] as patterns of muscle activity whose timing and/or amplitude modulation enable generation of different grasps. In the context of tendons, tension exerted on a tendon routed via multiple bodies generates torques at several joints (i.e., multi-articulation). These grippers utilise passive constant compliance, which means that the relationship between stiffness and force needs to made at design time. Variable compliance has the benefit that the gripper's joint stiffness can be adapted depending on the task and in real-time. For example, objects can be grasped adaptively with high compliance, and after stable grasp is achieved, stiffness can be increased to prevent the object from falling out of the gripper by acceleration or externally applied forces.

Variable stiffness can be achieved using control or adaptable inherent compliance [14]. Active impedance control exploits the control loop, which is flexible but does not allow energy storage and requires precise force control. Adjustable inherent compliance can be achieved in a variety of ways, including agonist/antagonist setups and variable transmission ratios between output and internal springs. We leverage an agonist-antagonist approach, which utilises two actuators in series with non-linear springs, driving a single joint. Adjustable stiffness is achieved by co-contraction of the nonlinear elements [15].

This work focuses on the design and control of an underactuated tendon-driven finger-based gripper with as key features high adaptability and variable joint stiffness (Fig. 1).

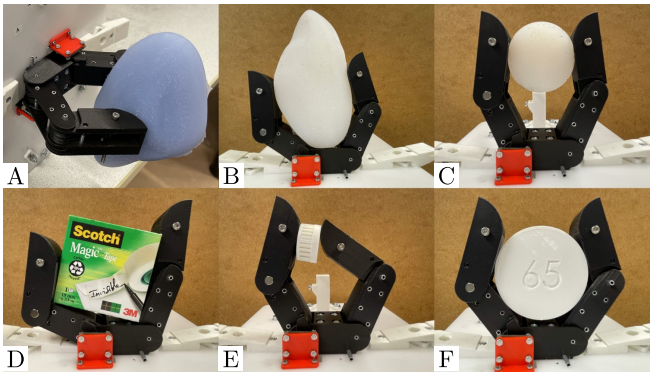


Fig. 1: Proof-of-concept prototype, grasping deformable and irregularly shaped objects with enveloping and pinch grasps: **(A-B)** chicken fillet mock-up (120 N/m), **(C)** deformable cylinder, **(D)** off-centre box, **(E)** bottle cap, and **(F)** large rigid cylinder. The video attachment shows various grasps in motion.

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The main contributions are summarised as follows:

- Kinematic analysis of tendon routing inside a finger-based gripper and their design parameters, in the context of grasp force and adjustable physical stiffness using co-contraction of non-linear springs;
- A control strategy to adjust joint stiffness independently from equilibrium positions and forces applied to the grasped object, as well control of different grasp types (envelope and pinch grasp, respectively);
- A proof-of-concept prototype with experimental results that demonstrate its key features.

The remainder is organised as follows. Sec. II will introduce the gripper concept. Sec. III focuses on modelling of the kinematics, force and stiffness properties, friction, and object interaction. The proposed control strategy is derived in Sec. IV. Sec. V presents the non-linear spring implementation and evaluation. Sec. VI presents the proof-of-concept prototype, followed by experimental results in Sec. VII. Finally, Sec. VIII and IX conclude the paper.

II. Concept and Analysis

Analysis of the human grip taxonomy [16], [17] showed that only two fingers are required to perform a parallel grasp, and that the other fingers wrap around an object to increase stability. Furthermore, a minimum of two phalanges per finger is required to achieve a 5-point stable grasp of a spherical object. Increasing the number of phalanges increases the number of stable gripper configurations, while also increasing complexity. For simplicity, the design described in this work is a 2-finger, 2 degree-of-freedom (DoF) per finger, planar gripper as shown in Fig. 2.

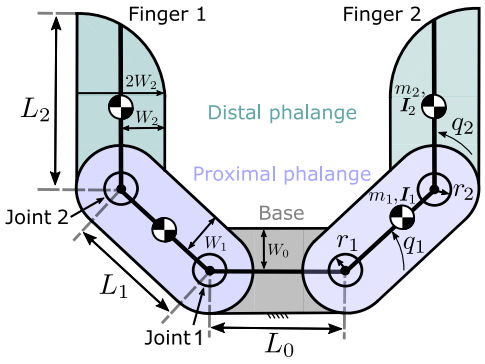


Fig. 2: Kinematic model of 2-DoF per finger gripper.

For a tendon-driven design, the tendons are routed around joint pulleys to apply joint torques. To enable grasping in tight spaces, such as from piles of objects, a compact design is desirable, with limited phalange width. As the moment arm of the tendons is given by the pulley radii, those radii are chosen nearly equal to the phalange width to reduce tendon forces. Furthermore, the tendon routing determines controllability, kinematic motion coupling, and bias forces [18]. Therefore, additional routing points guide the tendons to ensure contact with the pulleys. We utilise an optimal alternating pin allocation [19] resulting in a smooth contact force stabilisation as the fingers achieve a static equilibrium.

Joint pulleys and routing points can be implemented in different ways [20]; by 1) a tendon slipping over a fixed pulley, 2) a bushing-supported pulley or 3) a bearing-supported pulley. Each of these leads to different space requirements, complexity, and friction characteristics. In our design, we choose bearing-supported pulleys at the joints, and small-radius fixed pulleys for the routing points, primarily to achieve compactness.

A. Tendon routing and controlled variables

Many possible routings can be chosen for tendon-driven designs [18]. The 4-DoF gripper of Fig. 2 has 8 variables of interest; 4 joint angles and 4 corresponding stiffness values. The tendon routing configuration determines the actively and/or independently controlled variables. Achieving controllable joint stiffness requires co-contraction of two tendons, routed along both sides of the joint pulley. Conversely, actuating both tendons in opposite directions independently determines joint equilibrium. Enabling envelope and pinch grasps requires the equilibria of the two joints of each finger to be independently controlled.

Fig. 3 shows the chosen tendon routing. Tendons 1 and 2, that actuate joint 2 as agonist and antagonist, are routed via the inside of joint 1. Actuating tendons 1 and 2 in opposite directions does not affect joint 1, while co-contraction (i.e., increasing joint 2 stiffness) results in a closing force at joint 1. Conversely, tendon 3 opens joint 1. The result is that both joint equilibrium angles are independently controllable, while the stiffness values of both joints are coupled (detailed in Sec. III-C). To minimise the number of actuators, the gripper is symmetric, and joint angles and stiffness values of both fingers are coupled. The latter is achieved through an unactuated common mode of the two fingers, inducing high adaptability to object alignment and shape variation.

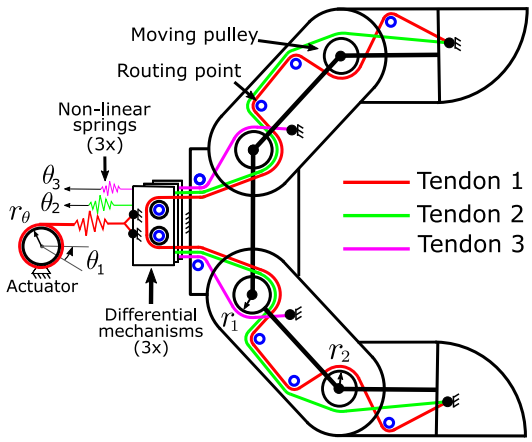


Fig. 3: Actuation and tendon routing.

The unactuated common mode coupling is achieved through a differential mechanism. As shown in Fig. 3, we utilise a double pulley differential [21], [22], owing to its large range of motion and straightforward integration of an encoder.

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III. Modelling

A. Generic kinematic model

To obtain the relation between tendon tension and joint torque, the kinematics of tendon-driven mechanisms are first derived, analogous to the derivation in [18]. We focus on a single finger, for the generic case with both passive and active tendons. In the following, L, P, N and A represent the number of tendons, passive tendons, joints and actuators, respectively. The relation between tendon elongations $\boldsymbol{l} \in \mathbb{R}^L$ and joint angles $\boldsymbol{q} = [q_1, q_2, \ldots, q_N]^\mathsf{T} \in \mathbb{R}^N$ and actuator configurations $\boldsymbol{\theta} = [\theta_1, \theta_2, \ldots, \theta_A]^\mathsf{T} \in \mathbb{R}^A$ is given by:

$$\boldsymbol{l}\left(\boldsymbol{q}, \boldsymbol{\theta}, \boldsymbol{l}_0\right) = \begin{bmatrix} \boldsymbol{l}_\mathrm{a}(\boldsymbol{q}, \boldsymbol{\theta}) \\ \boldsymbol{l}_\mathrm{p}\left(\boldsymbol{q}, \boldsymbol{l}_0\right) \end{bmatrix}, \tag{1}$$

where $\boldsymbol{l}_\mathrm{a}(\boldsymbol{q}, \boldsymbol{\theta}) \in \mathbb{R}^A$ and $\boldsymbol{l}_\mathrm{p}\left(\boldsymbol{q}, \boldsymbol{l}_0\right) \in \mathbb{R}^P$ denote the elongation of active and passive tendons, respectively. $\boldsymbol{l}_0 \in \mathbb{R}^P$ denotes the initial extension for the passive tendons. Defining $\boldsymbol{J}_\mathrm{t}$ to be a Jacobian matrix that maps joint velocities $\dot{\boldsymbol{q}}$ to tendon elongation rates $\dot{\boldsymbol{l}}$, the tendon elongation rates are given by:

$$\dot{\boldsymbol{l}} = \begin{bmatrix} \dot{\boldsymbol{l}}_\mathrm{a}(\boldsymbol{q}, \boldsymbol{\theta}) \\ \dot{\boldsymbol{l}}_\mathrm{p}(\boldsymbol{q}, \boldsymbol{l}_0) \end{bmatrix} = \boldsymbol{J}_\mathrm{t}\, \dot{\boldsymbol{q}} + \begin{bmatrix} \boldsymbol{R}_\theta \\ 0 \end{bmatrix} \dot{\boldsymbol{\theta}}, \tag{2}$$

with the constant Jacobian matrix

$$\boldsymbol{J}_\mathrm{t} = \begin{bmatrix} \boldsymbol{J}_\mathrm{a} \\ \boldsymbol{J}_\mathrm{p} \end{bmatrix}, \tag{3}$$

where $\boldsymbol{J}_\mathrm{a} \in \mathbb{R}^{(L-P)\times N}$ and $\boldsymbol{J}_\mathrm{p} \in \mathbb{R}^{P\times N}$ denote the active and passive Jacobian matrices, and $\boldsymbol{R}_\theta \in \mathbb{R}^{A\times A}$ is a diagonal matrix of which the elements are the radii of pulleys on the actuators. Using the principle of virtual work, the following expression for the joint torques $\boldsymbol{\tau}_\mathrm{t} \in \mathbb{R}^N$ is obtained:

$$\boldsymbol{\tau}_\mathrm{t} = -\boldsymbol{J}_\mathrm{t}^\mathsf{T}\, \boldsymbol{F}_\mathrm{t}, \tag{4}$$

where $\boldsymbol{F}_\mathrm{t} \in \mathbb{R}^L$ denotes the vector of tendon tension forces, for which holds $\boldsymbol{F}_\mathrm{t} \geq \boldsymbol{0}$, as the tendons cannot generate negative tension force.

B. Kinematic model of design

The configuration depicted in Fig. 3 has, per finger, no passive tendons ($P = 0$), three actuators ($A = 3$), three active tendons ($L = 3$), and two joints ($N = 2$), respectively. Note that for simplicity, we shall again consider a single finger. For the configuration in Fig. 3, $\boldsymbol{F}_\mathrm{t}$ is given by:

$$\boldsymbol{F}_\mathrm{t} = \begin{bmatrix} F_1 & F_2 & F_3 \end{bmatrix}^\mathsf{T}, \tag{5}$$

where F_1, F_2, F_3 denote tensile forces induced by deflection of the non-linear springs. The Jacobian $\boldsymbol{J}_\mathrm{t}$ is given by:

$$\boldsymbol{J}_\mathrm{t}^\mathsf{T} = \begin{bmatrix} -r_1 & -r_1 & r_1 \\ -r_2 & r_2 & 0 \end{bmatrix}. \tag{6}$$

Using Eq. (4), we obtain the joint torques as:

$$\boldsymbol{\tau}_\mathrm{t} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = -\begin{bmatrix} r_1\left(F_1 + F_2 - F_3\right) \\ r_2\left(F_1 - F_2\right) \end{bmatrix}. \tag{7}$$

To achieve the desired variable stiffness effect, the three springs in Fig. 3 are defined to have a quadratic force-displacement relation $F_i(l_i)$ for $i = 1, 2, 3$ given by:

$$F_i = \begin{cases} -a_i\left(l_i - l_0\right)^2 - b_i\left(l_i - l_0\right) - c & l_i \geq l_0, \\ 0 & \text{Otherwise.} \end{cases} \tag{8}$$

Here, a_i, b_i, c_i are design parameters, and the spring deflections $\boldsymbol{l} = [l_1, l_2, l_3]^\mathsf{T}$ are given by integrating Eq. (2):

$$\begin{aligned} l_1 &= r_a\theta_1 + r_1 q_1 + r_2 q_2, \\ l_2 &= r_a\theta_2 + r_1 q_1 - r_2 q_2, \\ l_3 &= r_a\theta_3 - r_1 q_1. \end{aligned} \tag{9}$$

At this point we take a specific choice for the coefficients, with the aim of obtaining a specific stiffness relation (Sec. III-C). Choosing F_1 and F_2 to have identical coefficients, that is, $a_1 = a_2 = a$, $b_1 = b_2 = b$, and $c_1 = c_2 = c$, and the coefficients of F_3 to be twice as large, i.e., $a_3 = 2a$, $b_3 = 2b$, and $c_3 = 2c$, the expressions for joint torques τ_1 and τ_2 reduce to:

$$\begin{aligned} \tau_1 = r_1[&ar_a^2(\theta_1^2 + \theta_2^2 - 2\theta_3^2) + 2ar_2^2 q_2^2 + \\ &2ar_1 r_a q_1(\theta_1 + \theta_2 + 2\theta_3) + 2ar_2 r_a q_2(\theta_1 - \theta_2) + \\ &br_a(\theta_1 + \theta_2 - 2\theta_3) + 4br_1 q_1], \\ \tau_2 = r_2[&ar_a^2(\theta_1^2 - \theta_2^2) + 2ar_a(r_1 q_1\theta_1 + r_2 q_2\theta_1 - r_1 q_1\theta_2 + \\ &r_2 q_2\theta_2) + 4ar_1 r_2 q_1 q_2 + 2br_2 q_2 + br_a(\theta_1 - \theta_2)]. \end{aligned} \tag{10}$$

C. Stiffness

Stiffness is defined as the rate of change of torque (or force) over (rotational) displacement:

$$k = \frac{\mathrm{d}\tau}{\mathrm{d}q}. \tag{11}$$

The instantaneous stiffness values of joints 1 and 2, k_1 and k_2 respectively, are then obtained by differentiating Eq. (10):

$$\begin{aligned} k_1 &= 2ar_1^2 r_a(\theta_1 + \theta_2 + 2\theta_3) + 4br_1^2, \\ k_2 &= 2ar_2^2 r_a(\theta_1 + \theta_2) + 4ar_1 r_2^2 q_1 + 2br_2^2. \end{aligned} \tag{12}$$

The specific choice of parameters made in Sec. III-B results in that both stiffness values are linearly dependent on co-contraction (i.e., the sum of θ_1 and θ_2, and θ_3 for k_1), with a constant offset proportional to b. Additionally, k_1 is independent of angle q_1. The offset b is desired to be non-zero, to prevent zero stiffness at zero spring deflection. This would have an effect similar to play and is therefore undesirable.

D. Friction model

Wrapping of the tendons around pulleys and routing points causes tensile force to decrease, between tensile force before, T_i, and tensile force after, T_{i+1}, with $T_i \geq T_{i+1}$. The force transmission ratio is denoted as $0 < \eta_i \leq 1$, such that $T_{i+1} = \eta_i T_i$. The value of η_i depends on the type of routing. η_i is given for slipping by [20]:

$$\eta_i = e^{-\mu\psi}, \tag{13}$$

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with friction coefficient μ and pulley contact angle ψ. For a bushing-supported or bearing-supported pulley, η_i is instead given by [20]:

$$\eta_i = \frac{\alpha^2 + \cos(\psi) - \sqrt{(1+\cos(\psi)\,(2\alpha^2 + \cos(\psi) - 1)}}{\alpha^2 - 1}. \tag{14}$$

with $\alpha = r_\mathrm{p}/(\mu r_\mathrm{j})$ for a journal bearing and with $\alpha = r_\mathrm{p}/(f_\mathrm{d} d)$ for a rolling bearing, where f_d is the friction moment coefficient similar to μ, and d is the diameter of the rolling bearing. Here we assume no slipping for bushing- and bearing supported pulleys, and no tendon stretching on the pulley surface.

These friction losses cause the tendon forces $\boldsymbol{F}_\mathrm{t}$ to not be perfectly transmitted to the distal pulleys, as Eq. (7) assumes. To account for this effect in our model, we define the torque caused by friction losses as $\boldsymbol{\tau}_\mathrm{f}$. Its formulation is straightforward and we omit it for brevity.

E. Contact forces and full dynamics

We model the contact along the continuous surface of the finger phalanges by a finite number of contact points, as shown in Fig. 4 (black dots). When in contact (red dots), a force is produced according to the object's stiffness. These forces produce generalised contact forces on the gripper, defined as:

$$\boldsymbol{\tau}_\mathrm{c} = \boldsymbol{J}_\mathrm{c}^\mathsf{T} \boldsymbol{F}_\mathrm{c}, \tag{15}$$

where $\boldsymbol{J}_\mathrm{c}$ denotes the Jacobian of all contact points in Fig. 4. The grasped object is modelled as a simple spherical stiffness, i.e., when in contact a linear spring with stiffness k_obj is connected between the centre-of-mass of the object and the contact point, producing linear contact forces $\boldsymbol{F}_\mathrm{c}$.

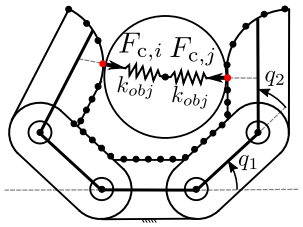


Fig. 4: Contact model. A finite number of contact points is distributed along the gripper surface (black dots). The deformable object is modelled as spherical spring.

Finally, the full dynamics of the gripper can be described in the well-known Euler-Lagrangian form, combining the applied forces by actuation, friction, and contact:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}_\mathrm{t} + \boldsymbol{\tau}_\mathrm{f} + \boldsymbol{\tau}_\mathrm{c}. \tag{16}$$

IV. Control

The goal of the control strategy is threefold; regulating 1) contact forces, 2) joint stiffness, and 3) grasp type. This is achieved using the actuation and contact Jacobians defined earlier, $\boldsymbol{J}_\mathrm{t}$ and $\boldsymbol{J}_\mathrm{c}$ respectively, together with known elongation-force relations of the nonlinear springs (8). Simultaneously, as was shown in Sec. III-C, co-contraction of opposing tendons can be utilised to regulate joint stiffness.

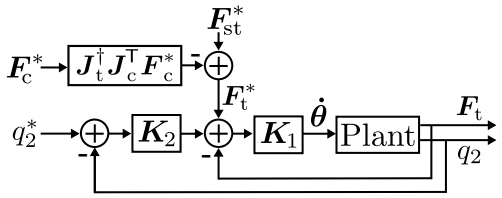


Fig. 5: Proposed control strategy.

The proposed control strategy is shown in Fig. 5, with velocity set points of the motors $\dot{\boldsymbol{\theta}}$ as inputs to the plant. It is comprised of two nested control loops, where the inner loop regulates tendon forces, and the outer loop regulates contact forces and joint stiffness. As a secondary control objective, the outer loop regulates the angle of joint 2 (q_2). This allows to regulate the angle of the contact forces, either 1) in parallel to achieve a pinch grasp, or 2) inward towards the gripper to move the object towards an enveloping/power grasp. Note that the plant in Fig. 5 includes the conversion from measured states (non-linear spring deflections and angular actuator positions) to control variables ($\boldsymbol{F}_\mathrm{a}$ and q_2) and low-level control loops for motor velocity control.

We start with regulation of contact forces and joint stiffness in the outer loop. Let C represent the number of contact points per finger. In equilibrium, we assume that the generalised forces produced by contacts and by actuation are equal, that is, $\boldsymbol{\tau}_\mathrm{t} + \boldsymbol{\tau}_\mathrm{c} = 0$ in (16), or $\boldsymbol{J}_\mathrm{t}^\mathsf{T}\boldsymbol{F}_\mathrm{t} = -\boldsymbol{J}_\mathrm{c}^\mathsf{T}\boldsymbol{F}_\mathrm{c}$. As the friction losses $\boldsymbol{\tau}_\mathrm{f}$ are very difficult to estimate accurately, they are neglected here. Provided a desired contact force $\boldsymbol{F}_\mathrm{c}^* \in \mathbb{R}^C$, this relation can be used to obtain desired tendon forces, $\boldsymbol{F}_\mathrm{t}^* \in \mathbb{R}^A$. Furthermore, due to the progressive stiffness of the elastic elements, an increase in joint stiffness can be achieved by increasing all tendon forces equally. Therefore, we introduce a bias force $\boldsymbol{F}_\mathrm{st}^* \in \mathbb{R}^A$, and the desired tendon forces $\boldsymbol{F}_\mathrm{t}^*$ are computed as

$$\boldsymbol{F}_\mathrm{t}^* = -(\boldsymbol{J}_\mathrm{t}^\mathsf{T})^\dagger \boldsymbol{J}_\mathrm{c}^\mathsf{T} \boldsymbol{F}_\mathrm{c}^* + \boldsymbol{F}_\mathrm{st}^*, \tag{17}$$

where $\boldsymbol{A}^\dagger := (\boldsymbol{A}^\mathsf{T}\boldsymbol{A})^{-1}\boldsymbol{A}^\mathsf{T}$ denotes the Moore-Penrose pseudo inverse of a matrix $\boldsymbol{A}$, since $\boldsymbol{J}_\mathrm{t}$ can be non-square, which it is in our case (Eq. (6)). The benefit of formulating the control objectives in terms of tendon forces is that no information on the object location or dimensions is necessary, as the motors will move until the desired contact forces are achieved.

The grasp type, enveloping or pinch grasp, is determined by setting the desired contact forces and joint 2 angle. Zero desired contact force at the proximal phalange results in a pinch grasp and a non-zero desired contact force on all phalanges results in an enveloping grasp. In addition, a bias force on the tendons along the second joint is introduced to control the joint 2 angle (q_2), with $\boldsymbol{K}_2 = \mathrm{diag}(K_2, -K_2, 0)$.

Finally, the spring deflections are controlled in the inner loop, with proportional gain $\boldsymbol{K}_1 = \mathrm{diag}(K_1, K_1, K_1)$. As the motors are velocity controlled, this provides asymptotic convergence to the desired tendon forces in a static configuration.

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A. State estimation with remote sensing

To compute the contact Jacobian $\boldsymbol{J}_c$, measurements or estimates of q_1 and q_2 are necessary. However, to keep complexity and size of the fingers to a minimum, we locate the sensor for measurement of joint 2 in the base. In the tendon system of Fig. 3, the configuration of joint 2 (q_2) can be reconstructed using the first joint's configuration q_1, displacement of the differential mechanism of tendon 1 ($r_\theta\theta_1 + l_1$), and displacement of that tendon within its differential mechanism, i.e., the passive DoF, which we denote by q_w. The equality describing lengths is given by:

$$r_1 q_1 + r_2 q_2 = r_\theta \theta_1 + l_1 - q_w. \tag{18}$$

Assuming equal joint radii ($r_1 = r_2 = r$), q_2 is given by:

$$q_2 = \frac{r_\theta \theta_1 + l_1 - q_w}{r} - q_1. \tag{19}$$

V. Non-linear spring implementation

The commercial availability of non-linear springs (NLS) with a specifiable force-displacement relationship is extremely limited. Therefore, (adapted) triangle mechanisms [23], four-bar mechanisms [24], and cam-followers [25] have been developed, to enable designed force-displacement behaviour in combination with linear springs. In this work, we choose a cam-follower, for its ease of implementation. To obtain the force-displacement relation, we follow the concept and solution proposed by [25][1]. The cam-follower design elongates linear springs using a roller guide following specific frame contour. The resultant force-deflection relationship is dependent on the contour shape.

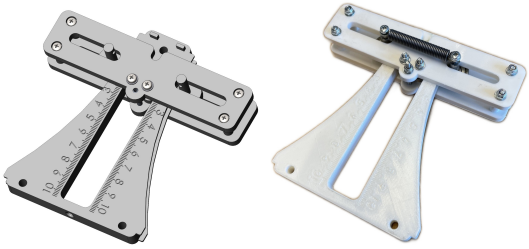


Fig. 6: Non-linear spring: CAD model and 3D printed prototype.

Fig. 6 shows the CAD model and 3D printed proof-of-concept. The design parameters were selected for the available components and to obtain a significant deflection range (of several cm), which reduces measurement errors/noise and limits the non-ideal behaviour of the linear springs. To prevent the roller guide from acting as a whippletree, additional (small) bearings are located on the roller guide on the inside profile, offset in the movement direction w.r.t. the outer bearings. Fig. 7 shows the force-deflection curves of three manufactured prototypes, together with a second-order fit and the desired curve. The data of all three devices demonstrate strong correlation to the desired quadratic behaviour (R^2 = 0.89, 0.88, and 0.86, respectively). The design parameters for the nonlinear springs were selected to achieve the relationships described in Sec. III-B and III-C, with maximum displacement and forces in line with the gripper prototype design presented in the next Section.

[1]Note that in [25], Eq. (12) should contain $\frac{x_0}{2}$ instead of x_0, and Eq. (13) should contain $y_c'(x_c)\left(\frac{F_p}{4k}\right)$ rather than $y_c'(x_c)^2\left(\frac{F_p}{4k}\right)$.

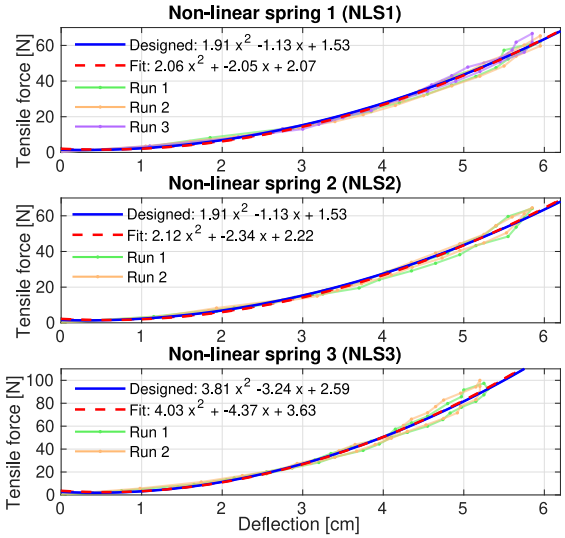


Fig. 7: Force-displacement relations of the non-linear springs.

VI. Prototype

The realised prototype is shown in Fig. 8. It comprises two compact fingers on a base which contains the non-linear springs, differential mechanism and actuators. All these parts were FDM-printed from PLA. The tendon guidance in the base is comprised of four bearing-guided nylon pulleys for each tendon with constant wrapping angle of 0.5π rad. The nylon finger joint pulleys are bushing-supported. Finally, the tendon routing pins (aluminium) in the phalanges are placed to ensure contact with the joints and smooth contact force stabilisation [19]. The tendons themselves are made from multiple-stranded polyethylene fibre (Spiderwire x8), that can be considered stiff under the applied forces.

The tendons are actuated using Dynamixel MX64 servo motors. Joint and pulley positions are measured using AMS AS5048A magnetic absolute position sensors. The geometric parameters are listed in Table I. The selected values enable 5-point stable enveloping and pinch grasps for deformable objects with diameter in the range 25-65 mm.

Parameter		Value	Unit
Actuator pulley radius	r_θ	7.5	mm
Joint pulley radii	r_1, r_2	7.5	mm
Base length	L_0	40.0	mm
Phalange lengths	L_1, L_2	47.5	mm
Phalange widths	W_1, W_2	10.0	mm

TABLE I: Prototype parameters.

The design parameters define the achievable joint stiffness as function of co-contraction force. Table II lists the theoretical joint stiffness values (Eq. (12)) for increasing co-contraction force (10 N to the maximum of 110 N). The ratio

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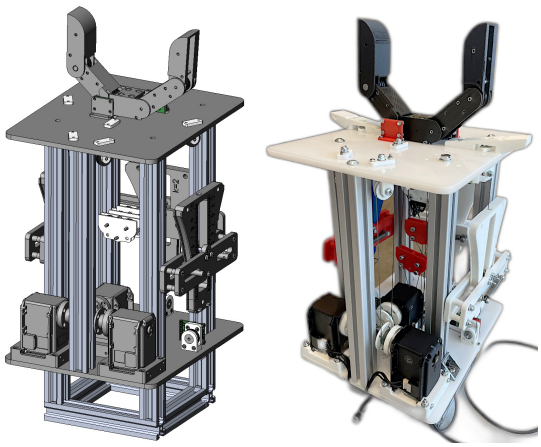


Fig. 8: CAD and proof-of-concept prototype gripper.

between stiffness of joint 1 and joint 2 is approximately a factor two (1.84-1.93) for the entire range.

Co-contraction force F_{st} [N]	**Joint 1 stiffness** k_1 [Nm/rad]	**Joint 2 stiffness** k_2 [Nm/rad]
10	0.46	0.25
35	0.81	0.43
60	1.04	0.54
85	1.23	0.64
110	1.39	0.72

TABLE II: Theoretical joint stiffness values.

VII. Results

We evaluate the key features and performance characteristics of the prototype through 3 experiments:

1) Object contact force regulation and joint stiffening;
2) Grasping (deformable) objects of various shapes/sizes;
3) Characterisation of tendon routing friction losses.

A. Experiment 1: Contact force regulation and stiffening

We first test the main feature of this design, which is to regulate contact forces independently from joint stiffness values. A deformable object is grasped with a desired contact force of 10 N, as shown in Fig. 9. Contact forces (bottom left figure) were measured using an instrumented object with ATI Mini40 force-torque sensor. After equilibrium is achieved, joint stiffness is increased through co-contraction (from 15 s to 25 s, top-right figure). Fig. 9 shows that a contact force of approx. 7.1 N (-29%) is achieved (bottom-left), due to frictional losses in tendon routing. During the experiment, the joint stiffness values (top-left) are calculated using Eq. (12), indicating the approximate increase in joint stiffness due to co-contraction. As intended, the contact force remains constant ($\pm 9\%$) when co-contraction increases the joint stiffness. Finally, as the grasp remains stable, joint angles remain approximately constant (bottom-right figure).

B. Experiment 2: Grasping different and off-centre objects

Figs. 10 and 11 show qualitative results of grasping objects that are both deformable and rigid, and vary in

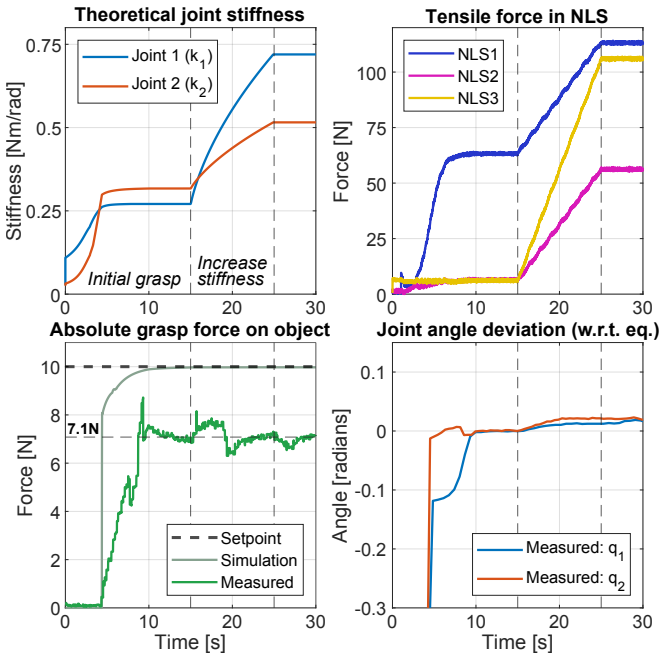


Fig. 9: Experiment 1: Grasping an object and stiffening joint 2.

shape, size, and mass. Fig. 10 shows grasping a 50 mm diameter deformable cylinder with stiffness of 120 N/m in both simulation and experiment. Although the free motion of the fingers differs slightly due to frictional effects in the prototype, the resulting grasp is similar. A reduced contact force can be observed in reduced squeezing of the real object, which can again be attributed to frictional effects[2]. Fig. 11 demonstrates both enveloping and pinch grasps of various objects, including off-centre due to the passive DoF shared by both fingers, while simultaneously regulating the contact force. Additionally, the **video attachment** shows various grasps in motion, including robust contact force regulation under externally applied forces and passive adjustability due to the unactuated common mode DoF of the fingers.

C. Experiment 3: Tendon routing friction losses

An important element in tendon-based designs is frictional losses. We characterise those of the prototype using measurements at four locations along the routing of tendon 1:

1) Immediately after the non-linear spring;
2) After two bearing guided pulleys with 'single winded' wrapping angle of 0.5π rad;
3) After two bearing guided pulleys with 'double winded' wrapping angle of 2.5π rad;
4) Tensile force at the joint pulley producing joint torque.

Fig. 12 shows the measurement setup. Measurements are produced using the Mini40 force-torque sensor connected using 3D printed brackets. The actuator of tendon 1 was driven to produce a ramped block wave force profile, between rest position and 90 N at the nonlinear spring. All measurements

[2]Note that the simulation does not visually show object deformation; therefore the fingers and object shape overlap.

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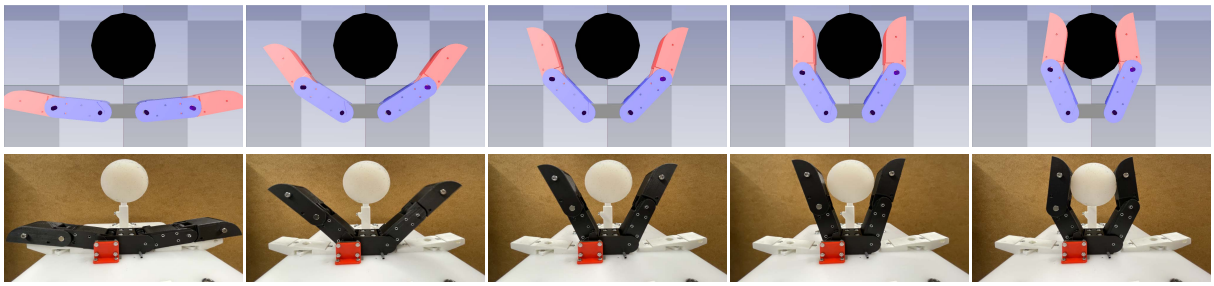


Fig. 10: Grasping a 50 mm diameter cylinder made from ECOFLEX 00-20 with stiffness of approx. 120 N/m.

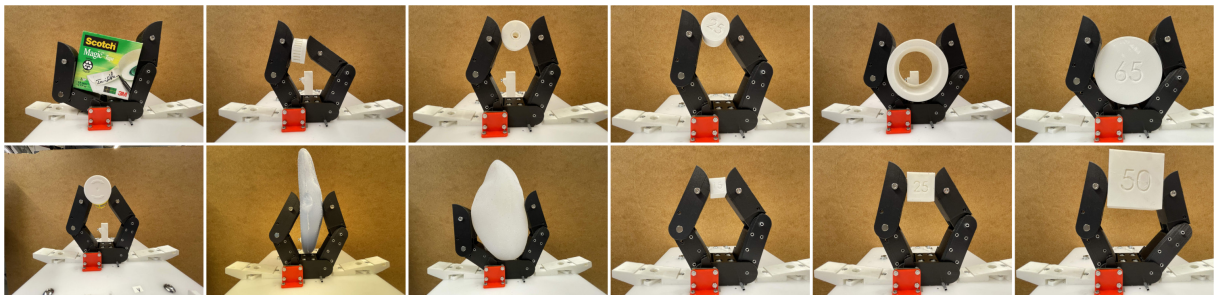


Fig. 11: Example grasp cases. The diameter or width is visualised in mm on the 3D printed items as a reference.

reflect cumulative friction losses along the tendon's routing up to the measurement location.

The results are shown in Fig. 13. They show that the cumulative friction losses can significantly reduce the joint torque. Compared to the force of the nonlinear spring (location 1), the average force routing efficiencies (defined in Sec. III-D) for positions 2, 3, and 4 are $\eta_2 = 0.72$, $\eta_3 = 0.61$, and $\eta_4 = 0.54$ respectively for 20 N, and up to $\eta_2 = 0.95$, $\eta_3 = 0.92$ and $\eta_4 = 0.90$ respectively for 90 N. Notably, although efficiency of single- and double-winded pulleys should be similar (Eq. (14)), η_3 is consistently lower than η_2. We attribute this to the neglected friction arising from the tendon moving past itself in the same v-groove of a double-winded pulley. Finally, all measurements show a slight creeping behaviour, which is typical for viscoelastic materials such as polymers.

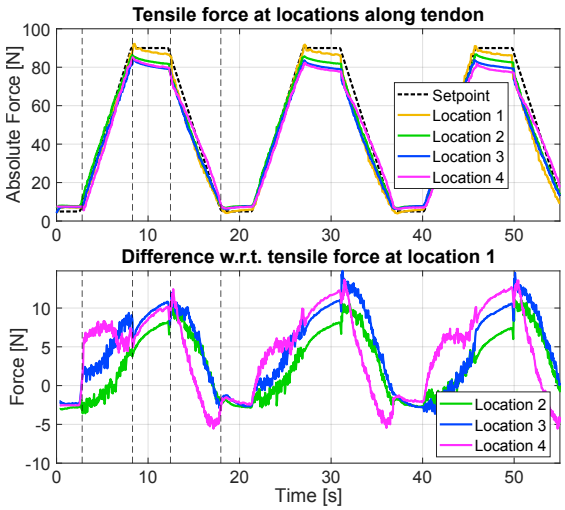


Fig. 13: Experiment 3: Tendon routing friction losses.

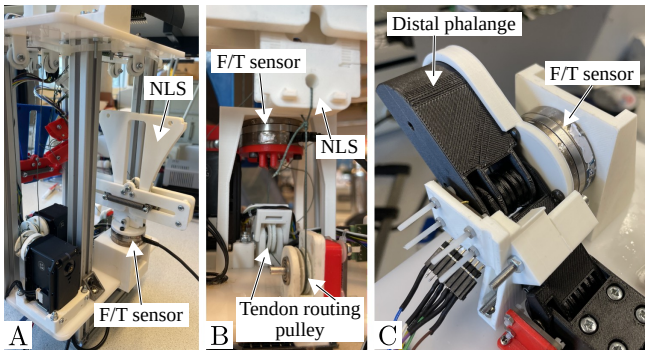


Fig. 12: Measurement setup of tendon tensile force: **(A)** Directly after non-linear spring (Location 1). **(B)** After two routing pulleys (Location 3). **(C)** At joint pulley (Location 4).

VIII. Discussion

The experimental results successfully validate the key features of the proposed gripper. Experiment 1 showed that contact force and joint stiffness values can be independently regulated. However, frictional losses in tendon routing result in reduced object contact forces compared to the set point. These losses were analysed in Experiment 3 and shown to be significant, especially for small contact forces (force transmission efficiency $\eta = 0.54$). Although this could be compensated for by increasing the contact force set point, we expect this to be configuration dependent, and therefore not straightforward. We consider it more appropriate to first make design changes, by reducing routing points, using only rolling contacts, and avoiding double v-groove crossings.

Experiment 2 demonstrated the ability to grasp objects

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that differ in shape, size, mass, and stiffness properties. Furthermore, demonstrated the ability of both enveloping and pinch grasp, and passive adaptability through the unactuated common-mode DoF. Although encouraging, these results are qualitative, and a more quantitative gripping performance analysis is needed.

IX. Conclusions and future work

This paper has presented the design and control of an underactuated tendon-driven finger-based gripper, with as key features high adaptability and variable joint stiffness.

We first analysed possible tendon routings, specifically focusing on joint torque and stiffness characteristics as function of nonlinear stiffness properties of the tendons. This led to a specific tendon routing topology and nonlinear stiffness parameters, enabling the controllability of joint equilibrium positions independently, coupled joint stiffness values, and an underactuated degree of freedom which enables passive adaptability to object position and shape. Subsequently a control strategy was proposed that leverages these features.

A proof-of-concept prototype was presented to validate the design principles and control approach. The results demonstrate successful regulation of object contact forces and joint stiffness values, independently from each other. Furthermore, the gripper qualitatively demonstrated the capability of grasping objects with a variety of shape, size, mass, and stiffness properties. Finally, the friction losses occurring in tendon routing were analysed.

Future work should focus on reducing tendon frictional effects. Subsequently, the realised joint stiffness values should be evaluated throughout the configuration space. Furthermore, gripping performance such as pull-out force should be evaluated systematically and related to joint stiffness. Towards practical use, the actuation and nonlinear stiffness elements should be more compactly integrated in the base, including the use of different materials, and the addition of more fingers should be considered for out-of-plane stability.

Acknowledgments

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